7.

The Cosmos on the Table

Trivial questions sometimes require deep and expansive knowledge of the cosmos just to answer them. In middle school chemistry class, I asked my teacher where the elements on the Periodic Table come from. He replied, Earth's crust. I'll grant him that. It's surely where the supply lab gets them. But how did Earth's crust acquire them? The answer must be astronomical. But in this case, do you actually need to know the origin and evolution of the universe to answer the question?

Yes, you do.

Only three of the naturally occurring elements were manufactured in the big bang. The rest were forged in the high-temperature hearts and explosive remains of dying stars, enabling subsequent generations of star systems to incorporate this enrichment, forming planets and, in our case, people.

For many, the Periodic Table of Chemical Elements is a forgotten oddity—a chart of boxes filled with mysterious, cryptic letters last encountered on the wall of high school chemistry class. As the organizing principle for the chemical behavior of all known and yet-to-be-discovered elements in the universe, the table instead ought to be a cultural icon, a testimony to the enterprise of science as an international human adventure conducted in laboratories, particle accelerators, and on the frontier of the cosmos itself.

Yet every now and then, even a scientist can't help thinking of the Periodic Table as a zoo of one-of-a-kind animals conceived by Dr. Seuss. How else could we believe that sodium is a poisonous, reactive metal that you can cut with a butter knife, while pure chlorine is a smelly, deadly gas, yet when added together they make sodium chloride, a